

Day 8 – Measuring Real Accuracy on Chapter 1

Evaluating My RAG System Properly, Instead of Trusting One Example · Souvik Pal · Personal Project

Objective

After Day 7's system was working, my mentor gave three directions: stop testing across all chapters and focus on one for accuracy, properly understand how the underlying vectors/embeddings work, and research which models or datasets are considered best in published work. Today's session covers the first task in depth: building a real, repeatable way to measure how accurate my system actually is on **Chapter 1: Chemical Reactions and Equations**, rather than judging it from a single question.

Building a Real Test Set

I wrote 12 exam-style questions covering the core concepts of Chapter 1 — chemical reactions, word equations, balanced equations, the four reaction types (combination, decomposition, displacement, double displacement), oxidation, reduction, corrosion, rancidity, and exothermic reactions — each paired with a correct reference answer. Each question was run automatically through the full system, searching only Chapter 1's content, and the AI's answer was compared against the reference answer.

Scoring Method 1 – Word-Overlap Matching

The first scoring method checked how many exact words the AI's answer shared with the reference answer (the same style of metric used as "Method 1" in my earlier Day 6 HotpotQA work). This gave a total accuracy of **50.0%** — which, on reading through the actual answers, looked wrong. Many of the AI's answers were clearly correct, just phrased with different words than my reference answer.

A real example that shows the problem:

Question: What is reduction?

AI's answer: "the loss of oxygen or gain of hydrogen"

Reference answer: "the addition of hydrogen or removal of oxygen"

These mean exactly the same thing, but share almost no exact words — so word-overlap scoring wrongly marked this as only "partial."

Scoring Method 2 – Meaning-Based Matching (Embeddings)

To fix this, I built a second scoring method using the same embedding technique that powers the search itself: convert both the AI's answer and the reference answer into meaning-codes

(embeddings), then measure how close those two codes are (called cosine similarity). This method judges whether two sentences mean the same thing, regardless of the exact words used.

Results – Both Scoring Methods Compared

Scoring Method	Accuracy	What it actually measures
Word-overlap matching	50.0%	Exact word matches only — misleading here
Meaning-based (embeddings)	95.8%	Whether the answer means the same thing — the true result

The two scoring methods only agreed with each other on 1 of the 12 questions. Reading through every answer manually confirmed the meaning-based score was the trustworthy one: 11 of 12 answers were genuinely correct, just worded differently than my reference answers.

The One Real Failure Found – "What is Oxidation?"

Both scoring methods agreed this one answer was weak. The AI replied that it could not determine the answer from the textbook content it was given. Rather than accepting that, I wrote a small script to inspect exactly which textbook passages the system had retrieved for this question.

The investigation showed the correct definition **was actually present** in the retrieved text ("if a substance gains oxygen or loses hydrogen ... it is oxidised"), but it was buried as the 4th of 5 retrieved passages, surrounded by unrelated exercise questions and some messy, duplicated text left over from the PDF extraction (e.g. repeated headings). The AI failed to notice the correct sentence within that clutter.

This is an important distinction: the **search** step succeeded (it found the right passage), but the **answer-writing** step failed to use it properly. This confirms the two stages of a RAG system can fail independently, and pinpoints a specific, fixable cause — messy extracted text — rather than a general weakness in the system.

Results Summary

Metric	Value
Test questions used	12 (Chapter 1 only)
Correct answers (meaning-based scoring)	11 out of 12 (95.8%)
Confirmed real failures	1 ("What is oxidation?")
Cause of the failure	Correct answer present but buried in cluttered retrieved text

What I Learned

- How you measure "correct" matters as much as the AI itself — a flawed scoring method made a good system look mediocre.
- Meaning-based (embedding) scoring is far more reliable than exact word-matching when judging free-text answers.
- "The AI couldn't find it" doesn't always mean the information was missing — sometimes it's present but hard to notice in cluttered text.
- Investigating one failure in depth is more valuable than reporting a single overall score.
- This mirrors the exact word-overlap limitation documented in my earlier Day 6 HotpotQA baseline (Method 1 vs. TF-IDF) — the same lesson reappeared in a completely different project.

Files Produced

- test_questions.json – 12 questions and reference answers for Chapter 1
- evaluate_chapter1.py – runs all questions, scores with both word-overlap and embedding similarity
- score_results.py – summarizes both scores side by side, with an agreement check
- debug_failed_questions.py – inspects the raw retrieved text behind any specific question
- evaluation_results.json – full saved results for all 12 questions, both scoring methods

Conclusion

Task 1 (focus on one chapter, measure real accuracy) is complete. My RAG system reaches 95.8% real accuracy on Chapter 1, confirmed through manual review, not just an automatic score. The one confirmed failure has a specific, understood cause — cluttered retrieved text hiding the correct sentence — rather than a fundamental flaw in the search or the model. This gives me a solid, evidence-backed foundation before moving on to the two remaining tasks my mentor set: explaining how vectors/embeddings work, and researching how my approach compares to published benchmarks and models.